Phase 5: Intelligent Emergency Triage Module

Development Report

LLMs for Emergency Triage and Clinical Decision Support
Internship Thesis Project | NED University of Engineering & Technology

1. Objective

Phase 5 focuses on designing and building the Intelligent Emergency Triage Module, the core decision-support component of the project. The module predicts a patient's Emergency Severity Index (ESI) level by combining two independent prediction sources: a baseline Machine Learning model trained on structured clinical features, and a fine-tuned Large Language Model (Qwen2.5-3B-Instruct, from Phase 4) that reasons over the patient's free-text clinical narrative. This report documents the module architecture, the development of both prediction components, and the integration and evaluation strategy used to combine them.

2. Dataset Overview

The dataset used in Phase 5 (**preprocessed_dataset.csv**) consists of 50 patient records with 65 features, generated during Phase 3 after preprocessing and standardization of clinical information. It contains structured vitals and demographics, chief complaint and clinical text, and expert-validated triage decisions.

Target variable: **acuity** (Emergency Severity Index, 1-5).

ESI Level	1	2	3	4	5
Patients	22	14	7	5	2

The class distribution is imbalanced, with more records at ESI 1-2 (higher acuity) than at ESI 3-5. This imbalance was factored into model evaluation choices in Section 4.

Evaluation ground truth: The **Final Decision** column marks expert-validated records (1 = RETAIN). Only the 36 RETAIN records are used for evaluating the ML model, the LLM, and the combined module, using **acuity** as the ground-truth ESI label.

3. Step 1 — Triage Module Architecture

Before implementation, the overall module design was defined, covering feature selection, the output format, and how the two models' predictions would be combined.

3.1 Structured Clinical Features (used by the ML model)

- Age, Temperature, Heart Rate, Respiratory Rate, Oxygen Saturation (SpO2)
- Systolic and Diastolic Blood Pressure, Pain Score
- Gender, Arrival Transport (walk-in / unknown)

Fields that only become available after the initial triage assessment — medications, procedures, ICU transfer, expert opinions, length of stay — were deliberately excluded to prevent data leakage.

3.2 Standardized Output Format

Every prediction produced by the module follows a fixed JSON structure so downstream evaluation and the future CDSS (Phase 6) can consume it consistently:

```
{
  "predicted_esi": 2, "ml_prediction": 2, "llm_prediction": 2,
  "agreement": true, "confidence": 0.94,
  "reason": "Patient has hypotension with severe chest pain...",
  "human_review": false
}
```

3.3 ML-LLM Decision Integration Strategy

- If both models predict the same ESI level → prediction accepted, agreement = true, human_review = false.
- If the predictions disagree → disagreement is recorded, the more urgent (lower ESI) prediction is preferred per project policy, agreement = false, human_review = true.

This keeps a healthcare professional in the loop for any case where the ML model and LLM disagree, consistent with the module being a decision-support tool rather than an autonomous decision-maker.

4. Feature Engineering & Baseline Machine Learning Model

Three baseline classifiers — Decision Tree, Random Forest, and XGBoost — were trained on the 11 structured features described in Section 3.1 to predict ESI level, and compared using Stratified 2-Fold Cross-Validation (chosen to fit the small, 50-record dataset while preserving class proportions across folds).

4.1 Model Comparison Results

Model	Accuracy	Precision	Recall	F1-Score
XGBoost	0.580	0.554	0.580	0.558
Random Forest	0.600	0.544	0.600	0.557
Decision Tree	0.480	0.482	0.480	0.463

Random Forest scored highest on accuracy, while XGBoost scored highest on weighted F1-score. Since triage is a safety-critical, imbalanced classification problem, F1-score was used as the primary selection metric, so **XGBoost** was chosen as the final baseline model.

4.2 Final Model

The selected XGBoost classifier was retrained on the full structured dataset and saved as **triage_ml_model.pkl**. A reload-and-predict check confirmed the saved model reproduces the expected prediction (predicted ESI = 1, matching the actual ESI = 1 on the test record).

4.3 Limitations

The model was trained on only 50 records (36 for evaluation), so performance estimates carry high variance and may not generalize to a real emergency department population. It is intended as a structured-data baseline to be combined with the LLM component, not a standalone clinical tool.

5. LLM-Based Triage Inference Module

This component builds the inference pipeline around the fine-tuned Qwen2.5-3B-Instruct model (with its Phase 4 LoRA adapter) to predict ESI level directly from a patient's free-text clinical narrative.

5.1 Pipeline

Patient Narrative → Prompt Template → Fine-Tuned Qwen Model (4-bit, LoRA adapter applied) → Generated Response → Parse ESI Level, Confidence, Reason → Structured JSON Output.

- **Prompting:** a fixed system prompt constrains the model to reply with JSON only ({"predicted_esi": <1-5>, "reason": "..."}), making downstream parsing reliable.
- **Generation:** greedy decoding (do_sample=False) is used for reproducible predictions, important for a clinical tool where the same input should always give the same output.
- **Confidence:** rather than an invented score, the token-level softmax probability of the generated ESI digit is extracted directly from the model's output distribution.

- **Parsing:** direct JSON parsing is attempted first, with a regex fallback for cases where the model adds stray text around the JSON. Unparseable output is returned as valid: false, rather than guessing or crashing.

5.2 Sample Output

Example on a synthetic chest-pain narrative:

```
{"predicted_esi": 2, "confidence": 0.9307, "reason": "Sudden onset crushing chest pain
radiating to the left arm, diaphoresis, and shortness of breath, with tachycardia,
hypotension, tachypnea and low SpO2, is highly suggestive of acute coronary syndrome...",
"valid": true}
```

The function was also tested on real expert-validated narratives from the dataset. Predictions were generally clinically reasonable, though not always matching the exact expert ESI level (e.g. predicting ESI 3 where the expert-assigned label was ESI 2 or 1 for two of the three sampled cases) — this is expected given the very small fine-tuning dataset, and is exactly the kind of gap the ML+LLM combination in Section 6 is designed to catch.

5.3 Deliverables

- **llm_triage_inference.py** — standalone inference module exposing predict_triage_llm(patient_text).
- **sample_predictions.ipynb** — this notebook, containing prompt design, parsing, confidence computation, and test cases.

6. Module Integration & Evaluation

The final stage combines Person A's ML model and Person B's LLM module into a single function, predict_triage(structured_data, patient_narrative), and evaluates the combined system against the 36 expert-validated records.

6.1 Integration Logic

Patient Information → [ML Prediction | LLM Prediction] (in parallel) → Decision Integration → Final Prediction. Decision integration applies the Section 3.3 policy: accept matching predictions, flag disagreements for human review and default to the more conservative (more urgent / lower ESI) prediction. The function additionally performs input validation and returns a combined confidence score alongside both individual model outputs, for example:

```
{"predicted_esi": 2, "ml_prediction": 3, "llm_prediction": 2, "agreement": false,
"confidence": 0.91, "reason": "Patient exhibits signs of cardiovascular instability."}
```

6.2 Evaluation Plan

The evaluation script scores the ML model, the LLM, and the combined system independently on the 36 RETAIN records, using acuity as ground truth, and reports Accuracy, Precision, Recall, F1-score, and a confusion matrix for the combined system. It further breaks results down into cases solved correctly by ML only, by LLM only, and by neither, to identify each component's strengths and weaknesses.

Note: the integration and evaluation notebook (integration.ipynb / triage_evaluation.ipynb) had not yet been run to completion at the time of this report — the pipeline and evaluation logic are implemented, but the resulting accuracy/precision/recall/F1 figures and confusion matrix are still pending a full run and should be added once available.

6.3 Deliverables

- **triage_module.py** — the integrated predict_triage() function.
- **integration.ipynb** — integration logic and worked example.
- **triage_evaluation.ipynb** — evaluation of ML, LLM, and combined system.
- **triage_evaluation_report.docx** — formal evaluation write-up (to be produced once results are available).

7. Summary

Phase 5 delivered a complete, working design for the Intelligent Emergency Triage Module: a documented architecture and integration policy, a baseline XGBoost model on structured features (weighted F1 = 0.558), a fine-tuned Qwen2.5-3B-Instruct inference pipeline with genuine token-level confidence scoring, and an integration layer that combines both with a human-review safeguard on disagreement. The module is now ready to be evaluated end-to-end and to feed into Phase 6 (Clinical Decision Support System).

7.1 Overall Limitations

- Very small dataset (50 records, 36 for evaluation) limits statistical confidence in all reported metrics.
- Class imbalance across ESI levels (few examples at ESI 4-5) may bias models toward higher-acuity classes.
- LLM predictions did not always match expert ESI labels exactly on the small validation sample — expected given limited fine-tuning data, and mitigated by the human-review flag on disagreement.
- Combined-system evaluation metrics are pending a completed run of the evaluation notebook.